Fiscal Year:	FY 2019	**Task Last Updated:**	FY 05/15/2020
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Project Title:	Label-free Photoacoustic Lymphatic Flowgraphy in Simulated Microgravity		
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Photoacoustic imaging can utilize endogenous optical absorption as the contrast to achieve structural and functional imaging. Its nonionizing and label-free features are desirable for spaceflight missions. We propose to develop label-free photoacoustic lymphatic flowgraphy (PALF) to image lymphatic flow and to study its alterations in simulated microgravity.
The specific aims are (1) to develop a spaceship compatible label-free PALF system and (2) to investigate the effect of microgravity on the alterations of cervical lymphatic flow and the involvement of such alterations in microgravity ocular syndrome (MOS) using label-free PALF.

A compact label-free PALF system with a single-element ultrasonic transducer will be developed. Laser beams will be

Task Description:

delivered through an optical fiber. Dry coupling will be used to eliminate the problem of using ultrasound gels in microgravity. By using an illumination wavelength of 975 nm, a peak of water absorption in the NIR (near infrared) optical window, water contents in the lymphatic vessels can be imaged. To measure the lymphatic flow velocity, an ultrasound encoded photoacoustic flowgraphy method combining ultrasound thermal tagging with photoacoustic imaging invented by our lab will be employed. PALF will be able to measure multiple key parameters of the lymphatic flows including the lymphatic vessel diameter, lymphatic flow direction, and flow velocity. PALF will be tested using both tissue phantoms and animals. In the phantom experiments, the accuracy of the flow velocity measurement will be examined by comparing with the preset flow velocities. In the animal experiments, the accuracy of the lymphatic flow velocity measurement in vivo will be validated by the results from invasive flow speed sensors. The system will also be applied to healthy human volunteers to demonstrate its capability in measuring human cervical lymphatic flows. The superficial lateral cervical lymph vessels of healthy human volunteers at the normal condition and at a spaceflight analog head down tilt bedrest will be imaged. The correlation of the cervical lymphatic flow changes and other gold-standard measures of physiological changes during bedrest will be studied.

To investigate the effect of microgravity on the alterations of cervical lymphatic flow and the involvement of such alterations in MOS, we plan to image two groups of rabbits, a control and an experimental group. The experimental group will be held in a head down tilt position for various durations. The cervical lymphatic flows of the rabbits in both groups will be monitored using the label-free PALF system throughout the experiments. After the experiments, key parameters related with MOS, including posterior globe flattening and optic disc protrusion, will be measured using magnetic resonance imaging and fundus examination. Any correlation between the alterations of the cervical lymphatic flow and the changes of the key parameters in MOS will be explored.

By developing PALF, we will enable label-free lymphatic flow imaging in simulated microgravity. PALF will be capable of measuring multiple key parameters of the lymphatic flow, providing a comprehensive evaluation of the lymphatic flow alteration in microgravity. PALF will have greater sensitivity in measuring slow flows in deep tissue than Doppler ultrasound flowmetry, currently used in the spaceflight. This is particularly important for characterizing the slow lymphatic flows in the low-pressure network. In the short term, PALF can serve as a test platform to study the lymphatic flow alterations and associated medical conditions in simulated microgravity on both experimental animals and human volunteers. In the long term, PALF has the potential to monitor lymphatic flows in astronauts during spaceflight missions and help to reduce or prevent the risks of spaceflight-associated medical conditions related with lymphatic flow alterations.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

INFORMATION FOR FINAL REPORTING NOVEMBER 2019

The lymphatic system is an essential part of both the immune and the circulatory systems. In the immune system, it removes invading microorganisms and defends against diseases. In the circulatory system, it maintains the interstitial fluid balance and absorbs fats and fat-soluble vitamins. It collects and returns excess interstitial fluid to blood in an open and low-pressure network. This characteristic makes the lymphatic flow vulnerable to microgravity, which may result in immune dysfunctions and fluid shifts in astronauts. The fluids shifts could possibly further cause ophthalmic changes including globe flattening, optic disc edema, and optic nerve protrusion, known as microgravity ocular syndromes (MOS). To prevent or reduce the risks of such spaceflight-associated medical conditions, it is critical to understand the effect of microgravity on lymphatic flow and the involvement of such an effect in MOS.

Medical imaging techniques serve as important tools to visualize the lymphatic system and measure lymphatic flows. Because lymphatic vessels contain few cells and carry mainly clear lymph, they are usually much more difficult to image than blood vessels. Most medical imaging techniques, including conventional lymphography, computed tomography lymphography, lymphoscintigraphy, and positron emission tomography, need exogenous contrast agents to help visualize lymphatic systems. However, injection of contrast agents is not suitable for use in spaceflight. Magnetic resonance lymphography can achieve imaging of the lymphatic system without contrast agents, but it is usually too bulky for spaceflight. Optical imaging techniques such as near-infrared fluorescence imaging have also been adopted for lymphography. They require injection of fluorescent tracers and suffer shallow penetration (due to the ~1 mm optical diffusion limit) for high-resolution imaging. The proposed photoacoustic lymphatic imaging technique will allow high-resolution imaging of lymphatic flow in vivo with a spaceship compatible system. To achieve compatibility for spaceflight, a compact photoacoustic imaging system will be developed. To measure the lymphatic flow velocity, an ultrasonically encoded photoacoustic flowgraphy method that combines ultrasonic tagging with photoacoustic imaging will be employed. With this method, PALF will have a higher sensitivity in measuring slow flows, particularly important for slow lymphatic flows in a low-pressured network, than the Doppler ultrasound flowmetry, currently used in spaceflight. With the proposed PALF system, multiple key parameters—including the lymphatic vessel diameter, flow direction and speed, and volumetric flow rate of the cervical lymphatic flow—can be measured. These key parameters will be monitored and compared among experimental groups with different exposure durations in simulated microgravity as well as with a control group in normal condition. A multidimensional comprehensive evaluation of cervical lymphatic flow alterations in simulated microgravity will be performed, which cannot be achieved by any other single medical imaging modality.

REPORTING FROM JANUARY 2019

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Medical imaging techniques serve as important tools to visualize the lymphatic system and measure lymphatic flows. Because lymphatic vessels contain few cells and carry mainly clear lymph, they are usually much more difficult to image than blood vessels. Most medical imaging techniques, including conventional lymphography, computed tomography lymphography, lymphoscintigraphy, and positron emission tomography need exogenous contrast agents to

	help visualize lymphatic systems. However, injection of contrast agents is not suitable for use in spaceflight. Magnetic resonance lymphography can achieve imaging of the lymphatic system without contrast agents, but it is usually too bulky for spaceflight. Ultrasound imaging techniques have been used in spaceships and International Space Stations (ISS). Ocular ultrasound was used to identify ophthalmic changes involved in MOS. However, to directly visualize the lymphatic system, microbubbles need to be injected to enhance contrast in ultrasound images. What's more, ultrasound imaging has very limited sensitivity in measuring slow flows such as lymphatic flows in the low-pressure network. Optical imaging techniques such as near-infrared fluorescence imaging have also been adopted for lymphography. They require injection of fluorescent tracers and suffer shallow penetration (due to the ~1 mm optical diffusion limit) for high-resolution imaging. The proposed photoacoustic lymphatic imaging technique, for the first time, will allow high-resolution label-free imaging of lymphatic flow in vivo with a spaceship compatible system. To achieve compatibility for spaceflight, a compact Pointing, Acquisition, and Tracking (PAT) system will be developed. By selecting an optimal illumination wavelength of 975 nm, an absorption peak of water in the near-infrared optical window, PALF can image the water content in lymphatic vessels without using any exogenous contrast agents. To measure the lymphatic flow velocity, an ultrasonically encoded photoacoustic flowgraphy method that combines ultrasonic tagging with photoacoustic imaging will be employed. With this method, PALF will have a higher sensitivity in measuring slow flows in deep tissue, particularly important for slow lymphatic flows in a low-pressured network, than Doppler ultrasound flowmetry, currently used in spaceflight. With the proposed PALF system, multiple key parameters, including the lymphatic vessel diameter, flow direction and speed, and volumetric flow rate, of the cervical lymphatic flow, can be measured. These key parameters will be monitored and compared among experimental groups with different exposure durations in simulated microgravity as well as with a control group in normal condition. A multidimensional comprehensive evaluation of cervical lymphatic flow alterations in simulated microgravity will be performed, which cannot be achieved by any other single medical imaging modality.
Task Progress:	INFORMATION FOR FINAL REPORTING NOVEMBER 2019 The original aim of this project for grant year 2 is to test and optimize the PALF system to measure the lymphatic flow in vivo. The proposed technique, if successfully developed, will enable label-free lymphatic flow imaging. By quantifying multiple key parameters of the lymphatic flow, it provides a comprehensive evaluation of the lymphatic flow alterations in microgravity. It will not only help to understand the contribution of microgravity-induced lymphatic flow alterations to microgravity ocular syndromes, but also serve as a test platform to study other spaceflight associated medical conditions in simulated microgravity. We have partially finished the tasks scheduled for the second grant year, as detailed below. • A photoacoustic lymphatic flowgraphy (PALF) system was optimized for in vivo small animal imaging. The PALF system can image mouse lymphatic vessels after injecting indocyanine green (ICG), an optical contrast agent with minimal toxicity. • Due to the low absorption of lymphatic fluid, PALF has difficulty measuring lymphatic flow without labeling. Therefore, we modified the current setup by utilizing a focused photoacoustic transducer to improve the sensitivity. The preliminary data shows promise. • As an alternative solution, we developed label-free ultrasonic flowgraphy. A focused ultrasonic transducer is used to heat a small region, altering the local speed of sound. This region is then tracked using ultrasonic imaging. Label-free tracking of saline flow has been demonstrated in phantoms. We are currently working on label-free imaging of the lymphatic flow in human subjects. • Using ultrasound imaging, we also captured changes in the size of lymph nodes and vessels. REPORTING FROM JANUARY 2019 The original aim of this project for grant year 1 is to develop a spaceship compatible label-free PALF system to measure multiple key parameters of the lymphatic flow. The proposed technique, if successfully developed, will enable label-free lymphatic flow imaging. By quantifying multiple key parameters of the lymphatic flow, it provides a comprehensive evaluation of the lymphatic flow alterations in microgravity. It will not only help to understand the contribution of microgravity-induced lymphatic flow alterations to microgravity ocular syndromes, but also serve as a test platform to study other spaceflight associated medical conditions in simulated microgravity. We have finished the tasks scheduled for the first grant year, as detailed below. a) Develop the PALF system operating at a 50 Hz frame rate. This PALF system can perform photoacoustic spectroscopy from 700 to 1000 nm, and provide a high spatial resolution in three dimensions: 85 micrometers, 260 micrometers, and 650 micrometers in the axial, lateral, and elevational directions, respectively. b) Test the PALF system using scattering tissue-mimicking phantoms, and validate the capacity of fluidic flow speed measurement, ranging from 0.5 to >10 mm/s. c) Verify that the PALF system conforms with both the laser and ultrasound safety standards. The former is governed by the American National Standards Institute (ANSI), the latter is given by the US Food and Drug Administration (FDA).
Bibliography Type:	Description: (Last Updated: 01/11/2019)
Awards	Wang L. "Michael S. Feld Biophotonics Award, The Optical Society, OSA, July 2018." Jul-2018
Awards	Wang L. "Induction to the National Academy of Engineering, July 2018." Jul-2018